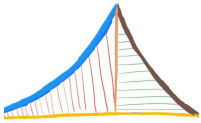
# Math for EE



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## About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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# 1. CLASS 12 BOARD PAPER

## A. Formulae

1.1.1. The *position vector* of the point $\mathbf{A}(x, y, z)$ is

$$\mathbf{a} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \tag{1.1.1.1}$$

1.1.2. The *magnitude* of $\mathbf{a}$ is

$$\|\mathbf{a}\| = \sqrt{\mathbf{a}^\top \mathbf{a}} = \sqrt{x^2 + y^2 + z^2} \tag{1.1.2.1}$$

1.1.3. The distance between the points $\mathbf{A}$ and $\mathbf{B}$ is

$$\|\mathbf{A} - \mathbf{B}\| = \sqrt{(\mathbf{A} - \mathbf{B})^\top (\mathbf{A} - \mathbf{B})} \tag{1.1.3.1}$$

1.1.4. The *unit vector* along $\mathbf{a} \neq \mathbf{0}$ is

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{\|\mathbf{a}\|} \tag{1.1.4.1}$$

1.1.5. The *section formula*: if $\mathbf{D}$ divides $AB$ in the ratio $k : 1$,

$$\mathbf{D} = \frac{k\mathbf{B} + \mathbf{A}}{k + 1} \tag{1.1.5.1}$$

1.1.6. The *dot product* and the angle $\theta$ between $\mathbf{a}$ and $\mathbf{b}$:

$$\mathbf{a}^\top \mathbf{b} = \|\mathbf{a}\| \, \|\mathbf{b}\| \cos\theta, \qquad \mathbf{a} \perp \mathbf{b} \iff \mathbf{a}^\top \mathbf{b} = 0 \tag{1.1.6.1}$$

1.1.7. The *cross product*:

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \, \|\mathbf{b}\| \sin\theta \tag{1.1.7.1}$$

The area of the triangle $ABC$ is

$$\frac{1}{2} \|(\mathbf{B} - \mathbf{A}) \times (\mathbf{C} - \mathbf{A})\| \tag{1.1.7.2}$$

1.1.8. The triangle inequality:

$$\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\| \tag{1.1.8.1}$$

1.1.9. The equation of a line through $\mathbf{A}$ with direction vector $\mathbf{m}$ is

$$\mathbf{x} = \mathbf{A} + \kappa\mathbf{m}, \quad \kappa \in \mathbb{R} \tag{1.1.9.1}$$

1.1.10. The plane with normal vector $\mathbf{n}$ can be expressed as

$$\mathbf{n}^\top \mathbf{x} = c \tag{1.1.10.1}$$

For the plane $ax + by + cz + d = 0$,

$$\mathbf{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \quad c = -d \tag{1.1.10.2}$$

The plane through $\mathbf{A}$ is $\mathbf{n}^\top (\mathbf{x} - \mathbf{A}) = 0$.

1.1.11. The distance of the point $\mathbf{P}$ from the plane $\mathbf{n}^\top \mathbf{x} = c$ is

$$\frac{\left|\mathbf{n}^\top \mathbf{P} - c\right|}{\|\mathbf{n}\|} \tag{1.1.11.1}$$

1.1.12. The planes $\mathbf{n}_1^\top \mathbf{x} = c_1$ and $\mathbf{n}_2^\top \mathbf{x} = c_2$ are parallel if

$$\mathbf{n}_1 = k\mathbf{n}_2 \tag{1.1.12.1}$$

1.1.13. The distance between the parallel planes $\mathbf{n}^\top \mathbf{x} = c_1$ and $\mathbf{n}^\top \mathbf{x} = c_2$ is

$$\frac{|c_1 - c_2|}{\|\mathbf{n}\|} \tag{1.1.13.1}$$

If the normals differ by a scale factor, first multiply one equation so that both have the same $\mathbf{n}$.

1.1.14. The *determinant* of $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and its inverse:

$$\left|\mathbf{A}\right| = ad - bc, \quad \mathbf{A}^{-1} = \frac{1}{\left|\mathbf{A}\right|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \tag{1.1.14.1}$$

1.1.15. The determinant of a $3 \times 3$ matrix, expanded along the first row, is

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a\,(ei - fh) - b\,(di - fg) + c\,(dh - eg) \tag{1.1.15.1}$$

If a row or column has a single nonzero entry, expand along it.

1.1.16. Properties of determinants:

$$\left|\mathbf{A}^\top\right| = \left|\mathbf{A}\right|, \quad \left|\mathbf{AB}\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\right|, \quad \left|\mathbf{A}^{-1}\right| = \frac{1}{\left|\mathbf{A}\right|} \tag{1.1.16.1}$$

1.1.17. The system $\mathbf{Ax} = \mathbf{b}$ with $\left|\mathbf{A}\right| \neq 0$ has the unique solution

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} \tag{1.1.17.1}$$

1.1.18. The *product* of $\mathbf{A} = \left(a_{ij}\right)_{m \times n}$ and $\mathbf{B} = \left(b_{ij}\right)_{n \times p}$ is

$$(\mathbf{AB})_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} \tag{1.1.18.1}$$

1.1.19. For $2 \times 2$ matrices,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} ap+br & aq+bs \\ cp+dr & cq+ds \end{pmatrix} \tag{1.1.19.1}$$

1.1.20. The *square* of a matrix is

$$\mathbf{A}^2 = \mathbf{A}\mathbf{A} \tag{1.1.20.1}$$

1.1.21. Basic derivatives:

$$\frac{d}{dx}x^n = nx^{n-1}, \quad \frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}\ln x = \frac{1}{x}, \tag{1.1.21.1}$$

$$\frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}\cos x = -\sin x \tag{1.1.21.2}$$

1.1.22. The product, quotient and chain rules:

$$(uv)' = u'v + uv', \quad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}, \quad \frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} \tag{1.1.22.1}$$

1.1.23. Basic integrals:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \;\; (n \neq -1), \quad \int \frac{dx}{x} = \ln|x| + C, \tag{1.1.23.1}$$

$$\int e^x\,dx = e^x + C, \quad \int \sin x\,dx = -\cos x + C, \quad \int \cos x\,dx = \sin x + C \tag{1.1.23.2}$$

1.1.24. The integrals

$$\int \sin kx\,dx = -\frac{\cos kx}{k} + C, \quad \int \cos kx\,dx = \frac{\sin kx}{k} + C \tag{1.1.24.1}$$

1.1.25. Integration by parts:

$$\int u\,dv = uv - \int v\,du \tag{1.1.25.1}$$

1.1.26. The fundamental theorem of calculus:

$$\int_a^b f(x)\,dx = F(b) - F(a), \quad F'(x) = f(x) \tag{1.1.26.1}$$

1.1.27. The double angle formulae:

$$\sin 2\theta = 2\sin\theta\cos\theta, \quad \cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 \tag{1.1.27.1}$$

1.1.28. By substitution $x = \sin\theta$, $\sqrt{1-x^2} = \cos\theta$ and

$$\tan\left(\sin^{-1}x\right) = \frac{x}{\sqrt{1-x^2}} \tag{1.1.28.1}$$

1.1.29. The first-order linear differential equation $\frac{dy}{dx} + Py = Q$ has the solution

$$y\, e^{\int P\, dx} = \int Q\, e^{\int P\, dx}\, dx + C \tag{1.1.29.1}$$

1.1.30. *Euler's method.* For $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$, with step size $h$,

$$y_{n+1} = y_n + hf(x_n, y_n), \quad x_{n+1} = x_n + h \tag{1.1.30.1}$$

1.1.31. A *linear programming problem* (LPP) optimises the *objective function*

$$Z = \mathbf{c}^\top \mathbf{x} \tag{1.1.31.1}$$

subject to the linear constraints $\mathbf{Ax} \leq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$.

1.1.32. The probability of an event $E$ and the addition rule:

$$\Pr(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}, \quad \Pr(A + B) = \Pr(A) + \Pr(B) - \Pr(AB) \tag{1.1.32.1}$$

1.1.33. For independent events $A$ and $B$,

$$\Pr(AB) = \Pr(A)\Pr(B), \quad \Pr(A + B) = 1 - \Pr(A')\Pr(B') \tag{1.1.33.1}$$

1.1.34. The conditional probability and Bayes' theorem:

$$\Pr(A|B) = \frac{\Pr(AB)}{\Pr(B)}, \quad \Pr(E_i|A) = \frac{\Pr(E_i)\Pr(A|E_i)}{\sum_j \Pr(E_j)\Pr(A|E_j)} \tag{1.1.34.1}$$

1.1.35. For a binomial random variable $X \sim B(n, p)$, $q = 1 - p$,

$$\Pr(X = r) = \binom{n}{r} p^r q^{n-r}, \quad \text{mean} = np, \quad \text{variance} = npq \tag{1.1.35.1}$$

1.1.36. A square matrix $\mathbf{A}$ is *singular* if

$$\left|\mathbf{A}\right| = 0 \tag{1.1.36.1}$$

1.1.37. A square matrix is *symmetric* if

$$\mathbf{A}^\top = \mathbf{A} \tag{1.1.37.1}$$

and *idempotent* if $\mathbf{A}^2 = \mathbf{A}$.

1.1.38. If $\mathbf{AB} = \mathbf{BA}$ (for example $\mathbf{B} = k\mathbf{I}$),

$$(\mathbf{A} + \mathbf{B})^3 = \mathbf{A}^3 + 3\mathbf{A}^2\mathbf{B} + 3\mathbf{AB}^2 + \mathbf{B}^3 \tag{1.1.38.1}$$

1.1.39. For $x > 0$,

$$\tan\left(\cot^{-1} x\right) = \frac{1}{x} \tag{1.1.39.1}$$

1.1.40. The tangent addition formula and the angle sum of a triangle:

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}, \quad A + B + C = \pi \tag{1.1.40.1}$$

1.1.41. The *integrating factor* of $\frac{dy}{dx} + Py = Q$ is

$$\mu(x) = e^{\int P\,dx}, \quad \mu' = P\mu \tag{1.1.41.1}$$

and $(\mu y)' = \mu Q$.

1.1.42. The differential equation $\frac{dy}{dx} = f(x)\,g(y)$ can be solved by separating the variables:

$$\int \frac{dy}{g(y)} = \int f(x)\,dx + C \tag{1.1.42.1}$$

1.1.43. For $k \neq 0$,

$$\int e^{kx}\,dx = \frac{e^{kx}}{k} + C \tag{1.1.43.1}$$

1.1.44. The second derivative of $y = \log x$ follows from $\frac{d}{dx}x^{-1} = -x^{-2}$:

$$\frac{dy}{dx} = \frac{1}{x} \implies \frac{d^2y}{dx^2} = -\frac{1}{x^2} \tag{1.1.44.1}$$

Numerically, with step size $h$,

$$y_n'' = \frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} \tag{1.1.44.2}$$

1.1.45. The *projection* of $\mathbf{A}$ on $\mathbf{B}$ is

$$\mathbf{C} = \frac{\mathbf{A}^\top \mathbf{B}}{\|\mathbf{B}\|^2}\mathbf{B}, \quad \|\mathbf{C}\| = \frac{\left|\mathbf{A}^\top \mathbf{B}\right|}{\|\mathbf{B}\|} \tag{1.1.45.1}$$

1.1.46. If $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ are unit vectors with angle $\theta$ between them,

$$\hat{\mathbf{a}}^\top \hat{\mathbf{a}} = \hat{\mathbf{b}}^\top \hat{\mathbf{b}} = 1, \quad \hat{\mathbf{a}}^\top \hat{\mathbf{b}} = \cos\theta \tag{1.1.46.1}$$

1.1.47. The *Laplace transform* of $f(x)$, $x \geq 0$, is

$$F(s) = \mathcal{L}\{f(x)\} = \int_0^\infty e^{-sx} f(x)\,dx \tag{1.1.47.1}$$

1.1.48. The Laplace transform of the derivative, and of $xf(x)$, are

$$\mathcal{L}\{f'(x)\} = sF(s) - f(0), \quad \mathcal{L}\{xf(x)\} = -\frac{dF(s)}{ds} \tag{1.1.48.1}$$

1.1.49. The standard transforms:

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{x\} = \frac{1}{s^2}, \quad \mathcal{L}\{e^{ax}\} = \frac{1}{s-a} \tag{1.1.49.1}$$

The inverse transform is found using partial fractions, for example

$$\frac{1}{s(s-a)} = \frac{1}{a}\left(\frac{1}{s-a} - \frac{1}{s}\right) \tag{1.1.49.2}$$

1.1.50. If $y_h$ is the solution of the homogeneous equation $\frac{dy}{dx} + Py = 0$, the integrating

factor of $\frac{dy}{dx} + Py = Q$ is

$$\mu = \frac{1}{y_h} = e^{\int P\,dx} \tag{1.1.50.1}$$

1.1.51. The *composition* of functions is $(f \circ g)(x) = f(g(x))$. The real cube root satisfies

$$\left(x^3\right)^{1/3} = x \quad \text{for all real } x \tag{1.1.51.1}$$

1.1.52. The triangle $ABC$ is right-angled at $C$ if

$$(\mathbf{A} - \mathbf{C})^{\top}(\mathbf{B} - \mathbf{C}) = 0 \iff \|\mathbf{A} - \mathbf{B}\|^2 = \|\mathbf{A} - \mathbf{C}\|^2 + \|\mathbf{B} - \mathbf{C}\|^2 \tag{1.1.52.1}$$

1.1.53. For $t > 0$ and $n > -1$,

$$\mathcal{L}\{t^n\} = \frac{\Gamma(n+1)}{s^{n+1}}, \quad \Gamma(n+1) = n\Gamma(n) \tag{1.1.53.1}$$

1.1.54. The *adjoint* of $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is

$$\operatorname{adj}\mathbf{A} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad \mathbf{A}(\operatorname{adj}\mathbf{A}) = (\operatorname{adj}\mathbf{A})\,\mathbf{A} = \left|\mathbf{A}\right|\mathbf{I} \tag{1.1.54.1}$$

1.1.55. The *z-transform* of the probability distribution $p[n] = \Pr(X = n)$ is

$$P(z) = \sum_{n} p[n]\, z^{-n} \tag{1.1.55.1}$$

The coefficient of $z^{-n}$ is $p[n]$, and

$$P(1) = 1, \quad \mathbb{E}[X] = -P'(1) \tag{1.1.55.2}$$

1.1.56. If $X_1$ and $X_2$ are independent, the distribution of $X_1 + X_2$ is the convolution of the two distributions, so its z-transform is the product

$$P_{X_1+X_2}(z) = P_{X_1}(z)\, P_{X_2}(z) \tag{1.1.56.1}$$

1.1.57. The Laplace transform of the logarithm, where $\gamma \approx 0.5772$ is Euler's constant:

$$\mathcal{L}\{\ln x\} = -\frac{\gamma + \ln s}{s} \tag{1.1.57.1}$$

1.1.58. The z-transform of the running sum $\sum_{m \le n} p[m]$, that is of $\Pr(X \le n)$, is

$$\frac{P(z)}{1 - z^{-1}} = P(z)\left(1 + z^{-1} + z^{-2} + \cdots\right) \tag{1.1.58.1}$$

so the coefficient of $z^{-n}$ is $\Pr(X \le n)$, and $\Pr(X \ge n+1) = 1 - \Pr(X \le n)$.

1.1.59. To maximise $Z = \mathbf{c}^{\top}\mathbf{x}$ subject to $\mathbf{A}\mathbf{x} \le \mathbf{b}$, $\mathbf{x} \ge \mathbf{0}$, the maximum occurs at a corner point of the feasible region. The corner point where the two lines $\mathbf{A}\mathbf{x} = \mathbf{b}$ meet is

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} \tag{1.1.59.1}$$

and a line $\mathbf{n}^{\top}\mathbf{x} = c$ meets the axes at $\left(\frac{c}{n_1}, 0\right)$ and $\left(0, \frac{c}{n_2}\right)$.

1.1.60. The distance between the skew lines $\mathbf{x} = \mathbf{A}_1 + \lambda\mathbf{m}_1$ and $\mathbf{x} = \mathbf{A}_2 + \mu\mathbf{m}_2$ is the length of the projection of $\mathbf{A}_2 - \mathbf{A}_1$ on the common perpendicular $\mathbf{n}$, where $\mathbf{n}$ solves

$$\begin{pmatrix} \mathbf{m}_1^\top \\ \mathbf{m}_2^\top \end{pmatrix} \mathbf{n} = \mathbf{0}, \qquad d = \frac{\left|\mathbf{n}^\top (\mathbf{A}_2 - \mathbf{A}_1)\right|}{\|\mathbf{n}\|} \tag{1.1.60.1}$$

The lines intersect iff they are not parallel and $d = 0$.

1.1.61. The planes $\mathbf{n}_1^\top \mathbf{x} = c_1$ and $\mathbf{n}_2^\top \mathbf{x} = c_2$ meet in a line, and every plane through that line is

$$(\mathbf{n}_1 + \lambda\mathbf{n}_2)^\top \mathbf{x} = c_1 + \lambda c_2 \tag{1.1.61.1}$$

1.1.62. Adding a multiple of one column (row) to another does not change a determinant. Such an operation is multiplication by an elementary matrix $\mathbf{E}$ with $\left|\mathbf{E}\right| = 1$, so

$$\left|\mathbf{ME}\right| = \left|\mathbf{EM}\right| = \left|\mathbf{M}\right| \tag{1.1.62.1}$$

A common factor of a column (row) can be taken out of the determinant.

1.1.63. The points $(x, y)$, $(x_1, y_1)$, $(x_2, y_2)$ are collinear, and the area of the triangle with vertices $(x_i, y_i)$, are given by

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0, \qquad \text{area} = \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right| \tag{1.1.63.1}$$

1.1.64. If $f'(x_0) = 0$, then $f$ has a maximum at $x_0$ if $f''(x_0) < 0$ and a minimum if $f''(x_0) > 0$. For a right circular cylinder of radius $r$ and height $h$,

$$\text{curved surface area} = 2\pi rh \tag{1.1.64.1}$$

1.1.65. The definite integral as a *limit of sums*, with $h = \frac{b-a}{n}$:

$$\int_a^b f(x)\, dx = \lim_{h\to 0}\ h\sum_{i=1}^{n} f(a + ih) \tag{1.1.65.1}$$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^{n} r^i = \frac{r(r^n - 1)}{r - 1} \tag{1.1.65.2}$$

1.1.66. *Partial fractions* with a repeated factor:

$$\frac{px + q}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b} \tag{1.1.66.1}$$

Comparing coefficients gives a linear system $\mathbf{M}\begin{pmatrix} A \\ B \\ C \end{pmatrix} = \mathbf{b}$, solved as $\begin{pmatrix} A \\ B \\ C \end{pmatrix} = \mathbf{M}^{-1}\mathbf{b}$.

1.1.67. *Logarithmic differentiation*: if $y = u^v$, then $\ln y = v \ln u$ and

$$\frac{dy}{dx} = u^v \left( v' \ln u + \frac{vu'}{u} \right) \tag{1.1.67.1}$$

1.1.68. The derivative of the inverse sine:

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1 - x^2}} \tag{1.1.68.1}$$

*B. Problems*

1.2.1. If $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$ are the position vectors of the points $\mathbf{A}(2, 3, -4)$, $\mathbf{B}(3, -4, -5)$ and $\mathbf{C}(3, 2, -3)$ respectively, then $|\mathbf{a} + \mathbf{b} + \mathbf{c}|$ is equal to

a) $\sqrt{113}$
b) $\sqrt{185}$
c) $\sqrt{203}$
d) $\sqrt{209}$

(CBSE 65-C-3, Class 12)

**Solution:** The position vectors are

$$\mathbf{a} = \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ -4 \\ -5 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 3 \\ 2 \\ -3 \end{pmatrix} \tag{1.2.1.1}$$

Adding them,

$$\mathbf{a} + \mathbf{b} + \mathbf{c} = \begin{pmatrix} 2 + 3 + 3 \\ 3 - 4 + 2 \\ -4 - 5 - 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 1 \\ -12 \end{pmatrix} \tag{1.2.1.2}$$

From (1.1.2.1),

$$\|\mathbf{a} + \mathbf{b} + \mathbf{c}\|^2 = \begin{pmatrix} 8 & 1 & -12 \end{pmatrix} \begin{pmatrix} 8 \\ 1 \\ -12 \end{pmatrix} = 64 + 1 + 144 = 209 \tag{1.2.1.3}$$

$$\implies \|\mathbf{a} + \mathbf{b} + \mathbf{c}\| = \sqrt{209} \tag{1.2.1.4}$$

Hence, the correct option is (d).

codes/q1.py

1.2.2. If $\mathbf{A} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, then $\mathbf{A}^2$ equals

a) $\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$
b) $\begin{pmatrix} 2 & -2 \\ -2 & -2 \end{pmatrix}$
c) $\begin{pmatrix} -2 & -2 \\ -2 & 2 \end{pmatrix}$
d) $\begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$

(CBSE 65-C-3, Class 12)

**Solution:** From (1.1.20.1) and (1.1.19.1),

$$\mathbf{A}^2 = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1+1 & -1-1 \\ -1-1 & 1+1 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix} \quad (1.2.2.1)$$

Hence, the correct option is (a).

codes/q2.py

1.2.3. The distance between the planes $4x - 4y + 2z + 5 = 0$ and $2x - 2y + z + 6 = 0$ is

a) $\frac{1}{6}$
b) $\frac{7}{6}$
c) $\frac{11}{6}$
d) $\frac{16}{6}$

(CBSE 65-C-3, Class 12)

**Solution:** From (1.1.10.1), the planes are $\mathbf{n}^\top \mathbf{x} = c$ with

$$\mathbf{n}_1 = \begin{pmatrix} 4 \\ -4 \\ 2 \end{pmatrix}, \ c_1 = -5; \quad \mathbf{n}_2 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}, \ c_2 = -6 \quad (1.2.3.1)$$

Since $\mathbf{n}_1 = 2\mathbf{n}_2$, the planes are parallel from (1.1.12.1). Scaling the second plane by 2,

$$\mathbf{n}_1^\top \mathbf{x} = -12 \quad (1.2.3.2)$$

From (1.1.13.1), the distance is

$$\frac{|c_1 - c_2'|}{\|\mathbf{n}_1\|} = \frac{|-5+12|}{\sqrt{16+16+4}} = \frac{7}{6} \quad (1.2.3.3)$$

Hence, the correct option is (b).

codes/q3.py

1.2.4. The probability of solving a specific question independently by $A$ and $B$ are $\frac{1}{3}$ and $\frac{1}{5}$ respectively. If both try to solve the question independently, the probability that the question is solved is

a) $\frac{7}{15}$
b) $\frac{8}{15}$
c) $\frac{2}{15}$
d) $\frac{14}{15}$

(CBSE 65-C-3, Class 12)

**Solution:** Let $A$ and $B$ be the events that the respective person solves the question. Then

$$\Pr(A) = \frac{1}{3}, \quad \Pr(B) = \frac{1}{5} \quad (1.2.4.1)$$

The question is solved unless both fail. From (1.1.33.1),

$$\Pr(A+B) = 1 - \Pr(A')\Pr(B') = 1 - \left(\frac{2}{3}\right)\left(\frac{4}{5}\right) = \frac{7}{15} \quad (1.2.4.2)$$

Hence, the correct option is (a).

codes/q4.py

1.2.5. The objective function of an LPP is

a) a constant
b) a linear function to be optimised
c) an inequality
d) a quadratic expression

(CBSE 65-C-3, Class 12)

**Solution:** From (1.1.31.1), the objective function is

$$Z = \mathbf{c}^\top \mathbf{x} = c_1 x + c_2 y \tag{1.2.5.1}$$

which is linear in $\mathbf{x}$ and is to be maximised or minimised subject to the constraints. Hence, the correct option is (b).

codes/q5.py

1.2.6. If $\sec^{-1}\left(\frac{1+x}{1-y}\right) = a$, then $\frac{dy}{dx}$ is equal to

a) $\frac{x-1}{y-1}$
b) $\frac{x-1}{y+1}$
c) $\frac{y-1}{x+1}$
d) $\frac{y+1}{x-1}$

(CBSE 65-C-3, Class 12)

**Solution:** Taking $\sec$ on both sides, with $k = \sec a$ a constant,

$$\frac{1+x}{1-y} = k \implies y = 1 - \frac{1+x}{k} \tag{1.2.6.1}$$

Differentiating with respect to $x$,

$$\frac{dy}{dx} = -\frac{1}{k} = -\frac{1-y}{1+x} = \frac{y-1}{x+1} \tag{1.2.6.2}$$

To verify numerically, take $a = 1$ and the initial point $\left(0,\, 1 - \frac{1}{k}\right)$ on (1.2.6.1). From (1.1.30.1), each option $\frac{dy}{dx} = f(x, y)$ is integrated with $h = 0.01$ and compared with (1.2.6.1). Only

$$f(x, y) = \frac{y-1}{x+1} \tag{1.2.6.3}$$

reproduces the curve; the other options deviate from it. Hence, the correct option is (c).

codes/q6.py

1.2.7. If $\int \frac{\cos 8x + 1}{\tan 2x - \cot 2x}\, dx = \lambda \cos 8x + c$, then the value of $\lambda$ is

a) $\frac{1}{16}$
b) $\frac{1}{8}$
c) $-\frac{1}{16}$
d) $-\frac{1}{8}$

(CBSE 65-C-3, Class 12)

**Solution:** From (1.1.27.1),

$$\tan 2x - \cot 2x = \frac{\sin^2 2x - \cos^2 2x}{\sin 2x \cos 2x} = \frac{-\cos 4x}{\frac{1}{2}\sin 4x} = -\frac{2\cos 4x}{\sin 4x} \tag{1.2.7.1}$$

$$\cos 8x + 1 = 2\cos^2 4x \tag{1.2.7.2}$$

Hence the integrand is

$$\frac{2\cos^2 4x \, \sin 4x}{-2\cos 4x} = -\sin 4x \cos 4x = -\frac{1}{2}\sin 8x \tag{1.2.7.3}$$

From (1.1.24.1),

$$\int -\frac{1}{2}\sin 8x \, dx = \frac{\cos 8x}{16} + c \implies \lambda = \frac{1}{16} \tag{1.2.7.4}$$

Hence, the correct option is (a).

codes/q7.py

1.2.9. $\int_0^1 \tan\left(\sin^{-1} x\right) \, dx$ equals

a) 2
b) 0
c) $-1$
d) 1

(CBSE 65-C-3, Class 12)

**Solution:** From (1.1.28.1),

$$\int_0^1 \tan\left(\sin^{-1} x\right) \, dx = \int_0^1 \frac{x}{\sqrt{1-x^2}} \, dx = \left[-\sqrt{1-x^2}\right]_0^1 = 0 - (-1) = 1 \tag{1.2.9.1}$$

Hence, the correct option is (d).

codes/q9.py

1.2.10. The roots of the equation $\begin{vmatrix} x & 0 & 8 \\ 4 & 1 & 3 \\ 2 & 0 & x \end{vmatrix} = 0$ are

a) $-4, \ 4$
b) $2, \ -4$
c) $2, \ 4$
d) $2, \ 8$

(CBSE 65-C-3, Class 12)

**Solution:** Expanding along the second column using (1.1.15.1),

$$\begin{vmatrix} x & 0 & 8 \\ 4 & 1 & 3 \\ 2 & 0 & x \end{vmatrix} = 1\left(x \cdot x - 8 \cdot 2\right) = x^2 - 16 = 0 \implies x = \pm 4 \tag{1.2.10.1}$$

Hence, the correct option is (a).

codes/q10.py

1.2.11. A square matrix $\mathbf{A}$ is said to be singular if _____.

**OR**

If $\mathbf{A} = \begin{pmatrix} 3 & -5 \\ 2 & 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 17 \\ 0 & -10 \end{pmatrix}$, then $\left|\mathbf{AB}\right| =$ _____. (CBSE 65-C-3, Class 12)

**Solution:** From (1.1.36.1), $\mathbf{A}$ is singular if $\left|\mathbf{A}\right| = 0$.
For the alternate question, from (1.1.16.1),

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right| \left|\mathbf{B}\right| \tag{1.2.11.1}$$

where

$$\left|\mathbf{A}\right| = 3\left(0\right) - \left(-5\right)\left(2\right) = 10, \quad \left|\mathbf{B}\right| = 1\left(-10\right) - 17\left(0\right) = -10 \tag{1.2.11.2}$$

$$\implies \left|\mathbf{AB}\right| = 10 \times \left(-10\right) = -100 \tag{1.2.11.3}$$

codes/q11.c

1.2.12. Two angles of a triangle are $\cot^{-1} 2$ and $\cot^{-1} 3$. The third angle of the triangle is _____. (CBSE 65-C-3, Class 12)

**Solution:** Let $A = \cot^{-1} 2$ and $B = \cot^{-1} 3$. From (1.1.39.1),

$$\tan A = \frac{1}{2}, \quad \tan B = \frac{1}{3} \tag{1.2.12.1}$$

From (1.1.40.1),

$$\tan\left(A + B\right) = \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}} = \frac{5/6}{5/6} = 1 \implies A + B = \frac{\pi}{4} \tag{1.2.12.2}$$

since $A, B$ are acute. The third angle is

$$C = \pi - \left(A + B\right) = \frac{3\pi}{4} \tag{1.2.12.3}$$

codes/q12.c

1.2.13. A card is picked at random from a pack of 52 playing cards. Given that the picked up card is a queen, the probability of it being a queen of spades is _____. (CBSE 65-C-3, Class 12)

**Solution:** Let $Q$ be the event that the card is a queen and $S$ the event that it is the queen of spades. Then

$$\Pr\left(Q\right) = \frac{4}{52}, \quad \Pr\left(S\right) = \frac{1}{52} \tag{1.2.13.1}$$

Since $S \subset Q$, $\Pr\left(SQ\right) = \Pr\left(S\right)$. From (1.1.34.1),

$$\Pr\left(S|Q\right) = \frac{\Pr\left(SQ\right)}{\Pr\left(Q\right)} = \frac{1/52}{4/52} = \frac{1}{4} \tag{1.2.13.2}$$

codes/q13.py

1.2.14. The integrating factor of the differential equation $x\frac{dy}{dx} - y = \log x$ is _____. (CBSE 65-C-3, Class 12)

**Solution:** Consider the homogeneous equation

$$x\frac{dy}{dx} - y = 0 \tag{1.2.14.1}$$

Let $Y(s) = \mathcal{L}\{y\}$. From (1.1.48.1),

$$\mathcal{L}\{xy'\} = -\frac{d}{ds}(sY - y(0)) = -Y - sY' \tag{1.2.14.2}$$

Taking the Laplace transform of the homogeneous equation,

$$-Y - sY' - Y = 0 \implies \frac{Y'}{Y} = -\frac{2}{s} \implies Y = \frac{C}{s^2} \tag{1.2.14.3}$$

From (1.1.49.1), the inverse transform is

$$y_h = Cx \tag{1.2.14.4}$$

From (1.1.50.1), the integrating factor is

$$\mu = \frac{1}{y_h} = \frac{1}{x} \tag{1.2.14.5}$$

This agrees with (1.1.41.1): dividing the equation by $x$ gives $P = -\frac{1}{x}$, so $\mu = e^{\int P\,dx} = e^{-\log x} = \frac{1}{x}$.

The code solves the homogeneous and the full equation using (1.1.30.1), and verifies $\mu = \frac{1}{x}$ and $(\mu y)' = \mu Q$.

codes/q14.py

1.2.15. If $y = \log x$, then $\frac{d^2y}{dx^2} =$ _____. (CBSE 65-C-3, Class 12)

**Solution:** Differentiating twice,

$$\frac{dy}{dx} = \frac{1}{x}, \quad \frac{d^2y}{dx^2} = -\frac{1}{x^2} \tag{1.2.15.1}$$

The result is verified numerically at $x = 2$ using (1.1.44.2).

codes/q15.py

1.2.16. If $\begin{pmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{pmatrix}$ is a symmetric matrix, then find the value of $x$.

**OR**

If $\mathbf{A}$ is a square matrix such that $\mathbf{A}^2 = \mathbf{A}$, then find $(2+\mathbf{A})^3 - 19\mathbf{A}$. (CBSE 65-C-3, Class 12)

**Solution:** From (1.1.37.1), the off-diagonal entries are equal,

$$x + 2 = 2x - 3 \implies x = 5 \tag{1.2.16.1}$$

For the alternate question, since $2\mathbf{I}$ commutes with $\mathbf{A}$, from (1.1.38.1),

$$(2\mathbf{I} + \mathbf{A})^3 = 8\mathbf{I} + 12\mathbf{A} + 6\mathbf{A}^2 + \mathbf{A}^3 \tag{1.2.16.2}$$

Using $\mathbf{A}^2 = \mathbf{A}$, $\mathbf{A}^3 = \mathbf{A}^2\mathbf{A} = \mathbf{A}$,

$$(2\mathbf{I} + \mathbf{A})^3 = 8\mathbf{I} + 19\mathbf{A} \implies (2 + \mathbf{A})^3 - 19\mathbf{A} = 8\mathbf{I} \tag{1.2.16.3}$$

codes/q16.py

1.2.17. Find the solution of the differential equation $\log\left(\frac{dy}{dx}\right) = ax + by$. (CBSE 65-C-3, Class 12)

**Solution:** The equation can be expressed as

$$\frac{dy}{dx} = e^{ax+by} = e^{ax}e^{by} \tag{1.2.17.1}$$

Let $u = e^{-by}$. Then

$$\frac{du}{dx} = -be^{-by}\frac{dy}{dx} = -be^{ax} \tag{1.2.17.2}$$

which is linear. Let $U(s) = \mathcal{L}\{u\}$. From (1.1.48.1) and (1.1.49.1),

$$sU - u(0) = -\frac{b}{s-a} \implies U = \frac{u(0)}{s} - \frac{b}{s(s-a)} \tag{1.2.17.3}$$

Using (1.1.49.2),

$$U = \frac{u(0) + \frac{b}{a}}{s} - \frac{b}{a}\cdot\frac{1}{s-a} \tag{1.2.17.4}$$

From (1.1.49.1), the inverse transform is

$$u = e^{-by} = u(0) + \frac{b}{a} - \frac{b}{a}e^{ax} \tag{1.2.17.5}$$

Dividing by $b$,

$$\frac{e^{ax}}{a} + \frac{e^{-by}}{b} = C \tag{1.2.17.6}$$

where $C = \frac{u(0)}{b} + \frac{1}{a}$ is a constant.
The code integrates the original equation using (1.1.30.1) and verifies that $e^{-by}$ matches the Laplace result and that the left hand side stays constant.

codes/q17.py

1.2.18. If $\mathbf{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\mathbf{b} = 5\hat{i} - 3\hat{j} - 4\hat{k}$, then find the ratio

$$\frac{\text{projection of vector } \mathbf{a} \text{ on vector } \mathbf{b}}{\text{projection of vector } \mathbf{b} \text{ on vector } \mathbf{a}}$$

**OR**

Let $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ be two unit vectors. If the vectors $\mathbf{c} = \hat{\mathbf{a}} + 2\hat{\mathbf{b}}$ and $\mathbf{d} = 5\hat{\mathbf{a}} - 4\hat{\mathbf{b}}$ are perpendicular to each other, then find the angle between the vectors $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$. (CBSE 65-C-3, Class 12)

**Solution:** The vectors are

$$\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ -3 \\ -4 \end{pmatrix} \tag{1.2.18.1}$$

$$\mathbf{a}^\top \mathbf{b} = 10 + 3 - 8 = 5, \quad \|\mathbf{a}\| = \sqrt{4+1+4} = 3, \quad \|\mathbf{b}\| = \sqrt{25+9+16} = 5\sqrt{2} \tag{1.2.18.2}$$

From (1.1.45.1), the required ratio is

$$\frac{\mathbf{a}^\top \mathbf{b}/\|\mathbf{b}\|}{\mathbf{a}^\top \mathbf{b}/\|\mathbf{a}\|} = \frac{\|\mathbf{a}\|}{\|\mathbf{b}\|} = \frac{3}{5\sqrt{2}} = \frac{3\sqrt{2}}{10} \tag{1.2.18.3}$$

For the alternate question, $\mathbf{c} \perp \mathbf{d}$ gives

$$\mathbf{c}^\top \mathbf{d} = \left(\hat{\mathbf{a}} + 2\hat{\mathbf{b}}\right)^\top \left(5\hat{\mathbf{a}} - 4\hat{\mathbf{b}}\right) = 5\hat{\mathbf{a}}^\top \hat{\mathbf{a}} + 6\hat{\mathbf{a}}^\top \hat{\mathbf{b}} - 8\hat{\mathbf{b}}^\top \hat{\mathbf{b}} = 0 \tag{1.2.18.4}$$

From (1.1.46.1),

$$5 + 6\cos\theta - 8 = 0 \implies \cos\theta = \frac{1}{2} \implies \theta = \frac{\pi}{3} \tag{1.2.18.5}$$

codes/q18.py

1.2.19. If $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = \left(3 - x^3\right)^{1/3}$, then find $(f \circ f)(x)$. (CBSE 65-C-3, Class 12)

**Solution:** From (1.1.51.1),

$$(f \circ f)(x) = f(f(x)) = \left(3 - (f(x))^3\right)^{1/3} \tag{1.2.19.1}$$

$$= \left(3 - \left(3 - x^3\right)\right)^{1/3} = \left(x^3\right)^{1/3} = x \tag{1.2.19.2}$$

So $f$ is its own inverse. The code checks this on $[-5, 5]$.

codes/q19.py

1.2.21. If $y = \sqrt{a + \sqrt{a+x}}$, then find $\frac{dy}{dx}$. (CBSE 65-C-3, Class 12)

**Solution:** The derivative of $t^n$, $n > 0$, is first obtained using the Laplace transform. Let $F(s) = \mathcal{L}\{t^n\}$. Since $t^n = 0$ at $t = 0$, from (1.1.48.1) and (1.1.53.1),

$$\mathcal{L}\left\{\frac{d}{dt}t^n\right\} = sF(s) = \frac{\Gamma(n+1)}{s^n} = n\frac{\Gamma(n)}{s^n} = n\,\mathcal{L}\left\{t^{n-1}\right\} \tag{1.2.21.1}$$

$$\implies \frac{d}{dt}t^n = nt^{n-1} \implies \frac{d}{dt}\sqrt{t} = \frac{1}{2\sqrt{t}} \tag{1.2.21.2}$$

Let $u = a + \sqrt{a+x}$, so that $y = \sqrt{u}$. From (1.2.21.2) with $t = a + x$ and $t = u$,

$$\frac{du}{dx} = \frac{1}{2\sqrt{a+x}}, \quad \frac{dy}{du} = \frac{1}{2\sqrt{u}} \tag{1.2.21.3}$$

From the chain rule (1.1.22.1),

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = \frac{1}{4\sqrt{a+x}\sqrt{a+\sqrt{a+x}}} \tag{1.2.21.4}$$

The code integrates this expression using (1.1.30.1) and checks that it recovers $y$.

codes/q21.py

1.2.22. Show that the three vectors $2\hat{i}-\hat{j}+\hat{k}$, $\hat{i}-3\hat{j}-5\hat{k}$ and $3\hat{i}-4\hat{j}-4\hat{k}$ form the vertices of a right-angled triangle. (CBSE 65-C-3, Class 12)

**Solution:** The vertices are

$$\mathbf{A} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 \\ -3 \\ -5 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 3 \\ -4 \\ -4 \end{pmatrix} \tag{1.2.22.1}$$

The sides are

$$\mathbf{B}-\mathbf{A} = \begin{pmatrix} -1 \\ -2 \\ -6 \end{pmatrix}, \quad \mathbf{C}-\mathbf{B} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{A}-\mathbf{C} = \begin{pmatrix} -1 \\ 3 \\ 5 \end{pmatrix} \tag{1.2.22.2}$$

From (1.1.2.1),

$$\|\mathbf{A}-\mathbf{B}\|^2 = 41, \quad \|\mathbf{B}-\mathbf{C}\|^2 = 6, \quad \|\mathbf{C}-\mathbf{A}\|^2 = 35 \tag{1.2.22.3}$$

Since $41 = 6+35$, from (1.1.52.1) the triangle is right-angled at $\mathbf{C}$. This is confirmed by

$$(\mathbf{A}-\mathbf{C})^\top (\mathbf{B}-\mathbf{C}) = \begin{pmatrix} -1 & 3 & 5 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} = 2+3-5 = 0 \tag{1.2.22.4}$$

codes/q22.py

1.2.23. A bag contains 19 tickets, numbered 1 to 19. A ticket is drawn at random and then another ticket is drawn without replacing the first one in the bag. Find the probability distribution of the number of even numbers on the ticket.

**OR**

Find the probability distribution of the number of successes in two tosses of a die, when a success is defined as "number greater than 5". (CBSE 65-C-3, Class 12)

**Solution:** The tickets 1 to 19 have 9 even and 10 odd numbers. Let $X$ be the number of even tickets, so $X \in \{0,1,2\}$. An outcome with $k$ even tickets contributes $z^{-k}$ to the z-transform (1.1.55.1), weighted by its probability. Following the two draws,

$$P(z) = \frac{9}{19}\left(\frac{8}{18}z^{-2} + \frac{10}{18}z^{-1}\right) + \frac{10}{19}\left(\frac{9}{18}z^{-1} + \frac{9}{18}\right) \tag{1.2.23.1}$$

$$= \frac{5}{19} + \frac{10}{19}z^{-1} + \frac{4}{19}z^{-2} \tag{1.2.23.2}$$

The inverse transform gives the coefficients of $z^{-n}$:

| $X=n$ | 0 | 1 | 2 |
|---|---|---|---|
| $p[n]$ | $\frac{5}{19}$ | $\frac{10}{19}$ | $\frac{4}{19}$ |

Check: $P(1)=\frac{5+10+4}{19}=1$, and $\mathbb{E}[X]=-P'(1)=\frac{10}{19}+\frac{8}{19}=\frac{18}{19}$.
For the alternate question, a single toss has success with probability $\frac{1}{6}$, so

$$Q(z)=\frac{5}{6}+\frac{1}{6}z^{-1} \tag{1.2.23.3}$$

The two tosses are independent, so from (1.1.56.1),

$$P(z)=Q(z)^2=\frac{25}{36}+\frac{10}{36}z^{-1}+\frac{1}{36}z^{-2} \tag{1.2.23.4}$$

| $X=n$ | 0 | 1 | 2 |
|---|---|---|---|
| $p[n]$ | $\frac{25}{36}$ | $\frac{10}{36}$ | $\frac{1}{36}$ |

Here $\mathbb{E}[X]=-P'(1)=\frac{10}{36}+\frac{2}{36}=\frac{1}{3}$.
The code simulates both experiments and compares the result with the theoretical distribution in Fig. 1.2.23.1.

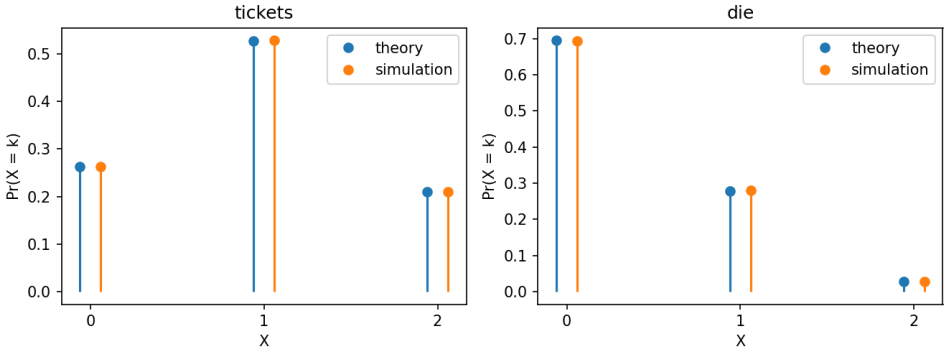



Fig. 1.2.23.1. Theoretical and simulated distributions

codes/q23.py

1.2.25. For the matrix $\mathbf{A}=\begin{pmatrix}2 & 3\\ -4 & -6\end{pmatrix}$, verify the following:

$$\mathbf{A}(\operatorname{adj}\mathbf{A})=(\operatorname{adj}\mathbf{A})\mathbf{A}=\left|\mathbf{A}\right|\mathbf{I}$$

(CBSE 65-C-3, Class 12)

**Solution:** From (1.1.54.1),

$$\operatorname{adj}\mathbf{A}=\begin{pmatrix}-6 & -3\\ 4 & 2\end{pmatrix},\quad \left|\mathbf{A}\right|=2(-6)-3(-4)=0 \tag{1.2.25.1}$$

Then

$$\mathbf{A}\left(\operatorname{adj}\mathbf{A}\right)=\begin{pmatrix}2&3\\-4&-6\end{pmatrix}\begin{pmatrix}-6&-3\\4&2\end{pmatrix}=\begin{pmatrix}-12+12&-6+6\\24-24&12-12\end{pmatrix}=\mathbf{0} \quad (1.2.25.2)$$

$$\left(\operatorname{adj}\mathbf{A}\right)\mathbf{A}=\begin{pmatrix}-6&-3\\4&2\end{pmatrix}\begin{pmatrix}2&3\\-4&-6\end{pmatrix}=\begin{pmatrix}-12+12&-18+18\\8-8&12-12\end{pmatrix}=\mathbf{0} \quad (1.2.25.3)$$

Both products equal $\mathbf{0}=\left|\mathbf{A}\right|\mathbf{I}$, which verifies the result.

codes/q25.py

1.2.26. Solve the following homogeneous differential equation:

$$x\frac{dy}{dx}=x+y$$

(CBSE 65-C-3, Class 12)

**Solution:** The equation can be expressed as

$$xy'-y=x \quad (1.2.26.1)$$

Let $Y\left(s\right)=\mathcal{L}\left\{y\right\}$. From (1.1.48.1), $\mathcal{L}\left\{xy'\right\}=-\left(Y+sY'\right)$, and from (1.1.49.1), $\mathcal{L}\left\{x\right\}=\frac{1}{s^2}$. Taking the Laplace transform,

$$-Y-sY'-Y=\frac{1}{s^2}\implies Y'+\frac{2}{s}Y=-\frac{1}{s^3} \quad (1.2.26.2)$$

The integrating factor is $e^{\int\frac{2}{s}ds}=s^2$, so

$$\left(s^2Y\right)'=-\frac{1}{s}\implies s^2Y=-\ln s+C\implies Y=\frac{C-\ln s}{s^2} \quad (1.2.26.3)$$

From (1.1.48.1) and (1.1.57.1),

$$\mathcal{L}\left\{x\ln x\right\}=-\frac{d}{ds}\left(-\frac{\gamma+\ln s}{s}\right)=\frac{1-\gamma-\ln s}{s^2} \quad (1.2.26.4)$$

$$\implies -\frac{\ln s}{s^2}=\mathcal{L}\left\{x\ln x\right\}-\frac{1-\gamma}{s^2} \quad (1.2.26.5)$$

Taking the inverse transform using (1.1.49.1),

$$y=x\ln x+\left(C-1+\gamma\right)x=x\left(\ln x+C_1\right) \quad (1.2.26.6)$$

where $C_1$ is a constant. Substituting back, $xy'=x\ln x+x+C_1x=x+y$.
The code integrates the equation using (1.1.30.1) for several values of $C_1$ and compares with $x\left(\ln x+C_1\right)$.

codes/q26.py

1.2.28. The random variable $X$ has a probability function $P\left(x\right)$ as defined below, where $k$

is some number:

$$P(x) = \begin{cases} k, & \text{if } x = 0 \\ 2k, & \text{if } x = 1 \\ 3k, & \text{if } x = 2 \\ 0, & \text{otherwise.} \end{cases}$$

Find:

(i) The value of $k$

(ii) $\Pr(X < 2)$, $\Pr(X \leq 2)$, $\Pr(X \geq 2)$

(CBSE 65-C-3, Class 12)

**Solution:** From (1.1.55.1), the z-transform of the distribution is

$$P(z) = k + 2kz^{-1} + 3kz^{-2} \tag{1.2.28.1}$$

(i) Since $P(1) = 1$,

$$k + 2k + 3k = 1 \implies k = \frac{1}{6} \tag{1.2.28.2}$$

$$\implies P(z) = \frac{1}{6}\left(1 + 2z^{-1} + 3z^{-2}\right) \tag{1.2.28.3}$$

(ii) From (1.1.58.1),

$$\frac{P(z)}{1 - z^{-1}} = \frac{1}{6}\left(1 + 2z^{-1} + 3z^{-2}\right)\left(1 + z^{-1} + z^{-2} + \cdots\right) = \frac{1}{6}\left(1 + 3z^{-1} + 6z^{-2} + 6z^{-3} + \right. \tag{1.2.28.4}$$

The coefficient of $z^{-n}$ is $\Pr(X \leq n)$. Hence

$$\Pr(X < 2) = \Pr(X \leq 1) = \frac{3}{6} = \frac{1}{2} \tag{1.2.28.5}$$

$$\Pr(X \leq 2) = \frac{6}{6} = 1 \tag{1.2.28.6}$$

$$\Pr(X \geq 2) = 1 - \Pr(X \leq 1) = \frac{1}{2} \tag{1.2.28.7}$$

The code compares the distribution with a simulation in Fig. 1.2.28.1.

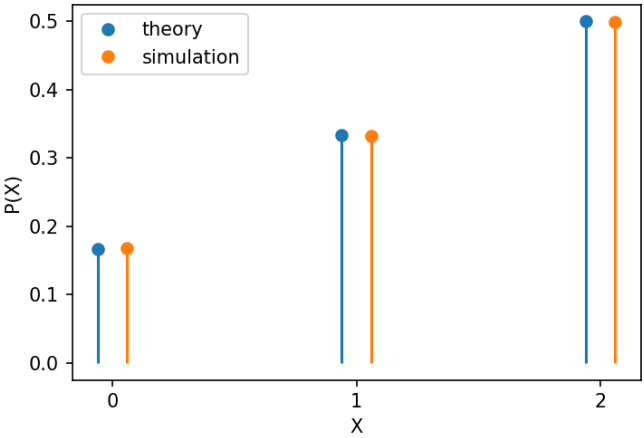



Fig. 1.2.28.1. Theoretical and simulated distribution

codes/q28.py

1.2.29. A company produces two types of goods, $A$ and $B$, that require gold and silver. Each unit of type $A$ requires 3 g of silver and 1 g of gold, while that of type $B$ requires 1 g of silver and 2 g of gold. The company can use at the most 9 g of silver and 8 g of gold. If each unit of type $A$ brings a profit of ₹ 120 and that of type $B$ ₹ 150, then find the number of units of each type that the company should produce to maximise profit. Formulate the above LPP and solve it graphically. Also, find the maximum profit. (CBSE 65-C-3, Class 12)

**Solution:** Let the company produce $x$ units of $A$ and $y$ units of $B$. From (1.1.31.1), the problem is to maximise

$$Z = \mathbf{c}^\top \mathbf{x}, \quad \mathbf{c} = \begin{pmatrix} 120 \\ 150 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \tag{1.2.29.1}$$

subject to

$$\mathbf{A}\mathbf{x} \le \mathbf{b}, \quad \mathbf{x} \ge \mathbf{0}, \quad \mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 9 \\ 8 \end{pmatrix} \tag{1.2.29.2}$$

that is, $3x + y \le 9$ (silver) and $x + 2y \le 8$ (gold). The corner points of the feasible region follow from (1.1.59.1). The lines meet at

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 9 \\ 8 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \tag{1.2.29.3}$$

The silver line meets the $x$-axis at $(3, 0)$ and the gold line meets the $y$-axis at $(0, 4)$. The other two intercepts, $(0, 9)$ and $(8, 0)$, violate the other constraint. Evaluating $Z = \mathbf{c}^\top \mathbf{x}$ at the corner points,

| Corner point | $(0,0)$ | $(3,0)$ | $(2,3)$ | $(0,4)$ |
|---|---|---|---|---|
| $Z$ | 0 | 360 | 690 | 600 |

The maximum is at $(2, 3)$. The company should produce 2 units of type $A$ and 3 units of type $B$, for a maximum profit of ₹ 690.

codes/q29.py

1.2.30. Find the shortest distance between the following lines and hence write whether the lines are intersecting or not.

$$\frac{x-1}{2} = \frac{y+1}{3} = z, \qquad \frac{x+1}{5} = \frac{y-2}{1},\ z = 2$$

**OR**

Find the equation of the plane through the line of intersection of the planes $\mathbf{r} \cdot \left(\hat{i} + 3\hat{j}\right) + 6 = 0$ and $\mathbf{r} \cdot \left(3\hat{i} - \hat{j} - 4\hat{k}\right) = 0$, which is at a unit distance from the origin. (CBSE 65-C-3, Class 12)

**Solution:** A line through $\mathbf{A}$ with direction $\mathbf{m}$ is $\mathbf{x} = \mathbf{A} + \lambda\mathbf{m}$. The two lines are

$$\mathbf{A}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix},\ \mathbf{m}_1 = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}; \qquad \mathbf{A}_2 = \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix},\ \mathbf{m}_2 = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} \tag{1.2.30.1}$$

Since $\mathbf{m}_1 \neq k\mathbf{m}_2$, the lines are not parallel. A vector $\mathbf{n}$ perpendicular to both lines satisfies (1.1.6.1),

$$\begin{pmatrix} \mathbf{m}_1^\top \\ \mathbf{m}_2^\top \end{pmatrix} \mathbf{n} = \begin{pmatrix} 2 & 3 & 1 \\ 5 & 1 & 0 \end{pmatrix} \mathbf{n} = \mathbf{0} \tag{1.2.30.2}$$

From the second row $n_2 = -5n_1$, and from the first row $n_3 = -2n_1 - 3n_2 = 13n_1$. Taking $n_1 = -1$,

$$\mathbf{n} = \begin{pmatrix} -1 \\ 5 \\ -13 \end{pmatrix}, \qquad \|\mathbf{n}\|^2 = 1 + 25 + 169 = 195 \tag{1.2.30.3}$$

From (1.1.60.1), the shortest distance is the length of the projection of $\mathbf{A}_2 - \mathbf{A}_1$ on $\mathbf{n}$, using (1.1.45.1),

$$\mathbf{A}_2 - \mathbf{A}_1 = \begin{pmatrix} -2 \\ 3 \\ 2 \end{pmatrix}, \qquad d = \frac{\left|\mathbf{n}^\top (\mathbf{A}_2 - \mathbf{A}_1)\right|}{\|\mathbf{n}\|} = \frac{|2 + 15 - 26|}{\sqrt{195}} = \frac{9}{\sqrt{195}} \tag{1.2.30.4}$$

Since $d \approx 0.64 \neq 0$, the lines do **not** intersect (they are skew).
For the alternate question, from (1.1.10.1) the two planes are

$$\mathbf{n}_1^\top \mathbf{x} = c_1,\ \mathbf{n}_1 = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix},\ c_1 = -6; \qquad \mathbf{n}_2^\top \mathbf{x} = c_2,\ \mathbf{n}_2 = \begin{pmatrix} 3 \\ -1 \\ -4 \end{pmatrix},\ c_2 = 0 \tag{1.2.30.5}$$

From (1.1.61.1), every plane through their line of intersection is

$$\left(\mathbf{n}_1 + \lambda\mathbf{n}_2\right)^\top \mathbf{x} = c_1 + \lambda c_2 = -6 \tag{1.2.30.6}$$

Its distance from the origin is $\frac{6}{\|\mathbf{n}_1+\lambda\mathbf{n}_2\|}=1$. Since $\mathbf{n}_1^\top\mathbf{n}_2=3-3=0$,

$$\|\mathbf{n}_1+\lambda\mathbf{n}_2\|^2=10+26\lambda^2=36 \implies \lambda=\pm1 \tag{1.2.30.7}$$

$$\lambda=1:\ \begin{pmatrix}4\\2\\-4\end{pmatrix}^\top\mathbf{x}=-6, \qquad \lambda=-1:\ \begin{pmatrix}-2\\4\\4\end{pmatrix}^\top\mathbf{x}=-6 \tag{1.2.30.8}$$

Hence the planes are

$$2x+y-2z+3=0 \quad \text{or} \quad x-2y-2z-3=0 \tag{1.2.30.9}$$

each at distance $\frac{3}{3}=1$ from the origin.

codes/q30.py

1.2.31. Find a unit vector perpendicular to each of the vectors $(\mathbf{a}+\mathbf{b})$ and $(\mathbf{a}-\mathbf{b})$, where $\mathbf{a}=\hat{i}+\hat{j}+\hat{k}$ and $\mathbf{b}=\hat{i}+2\hat{j}+3\hat{k}$. (CBSE 65-C-3, Class 12)

**Solution:** The two vectors are

$$\mathbf{p}=\mathbf{a}+\mathbf{b}=\begin{pmatrix}2\\3\\4\end{pmatrix}, \qquad \mathbf{q}=\mathbf{a}-\mathbf{b}=\begin{pmatrix}0\\-1\\-2\end{pmatrix} \tag{1.2.31.1}$$

A vector $\mathbf{n}$ perpendicular to both satisfies (1.1.6.1),

$$\begin{pmatrix}\mathbf{p}^\top\\\mathbf{q}^\top\end{pmatrix}\mathbf{n}=\begin{pmatrix}2&3&4\\0&-1&-2\end{pmatrix}\mathbf{n}=\mathbf{0} \tag{1.2.31.2}$$

The second row gives $n_2=-2n_3$ and the first row gives $2n_1=-3n_2-4n_3=2n_3$, so $n_1=n_3$. Taking $n_3=-1$,

$$\mathbf{n}=\begin{pmatrix}-1\\2\\-1\end{pmatrix}, \qquad \|\mathbf{n}\|^2=1+4+1=6 \tag{1.2.31.3}$$

Dividing by the norm, the unit vectors perpendicular to both $\mathbf{p}$ and $\mathbf{q}$ are

$$\hat{\mathbf{n}}=\pm\frac{\mathbf{n}}{\|\mathbf{n}\|}=\pm\frac{1}{\sqrt{6}}\begin{pmatrix}-1\\2\\-1\end{pmatrix} \tag{1.2.31.4}$$

that is, $\pm\frac{1}{\sqrt{6}}\left(-\hat{i}+2\hat{j}-\hat{k}\right)$. Check: $\mathbf{n}^\top\mathbf{p}=-2+6-4=0$ and $\mathbf{n}^\top\mathbf{q}=-2+2=0$.

codes/q31.py

1.2.32. Using properties of determinants show that

$$\begin{vmatrix}1+a^2-b^2 & 2ab & -2b\\ 2ab & 1-a^2+b^2 & 2a\\ 2b & -2a & 1-a^2-b^2\end{vmatrix}=\left(1+a^2+b^2\right)^3.$$

**OR**

Find the equation of the line joining $A\,(1,3)$ and $B\,(0,0)$ using determinants. Also,

find $k$ if $D\left(k,0\right)$ is a point such that the area of $\triangle ABD$ is 3 square units. (CBSE 65-C-3, Class 12)

**Solution:** Let $\mathbf{M}$ be the given matrix and $s=1+a^2+b^2$. The column operations $C_1\to C_1-bC_3$ and $C_2\to C_2+aC_3$ are right multiplication by

$$\mathbf{E}_1=\begin{pmatrix}1&0&0\\0&1&0\\-b&0&1\end{pmatrix},\quad \mathbf{E}_2=\begin{pmatrix}1&0&0\\0&1&0\\0&a&1\end{pmatrix},\quad \left|\mathbf{E}_1\right|=\left|\mathbf{E}_2\right|=1 \tag{1.2.32.1}$$

From (1.1.62.1) and (1.1.16.1), $\left|\mathbf{M}\right|=\left|\mathbf{M}\mathbf{E}_1\mathbf{E}_2\right|$, and

$$\mathbf{M}\mathbf{E}_1\mathbf{E}_2=\begin{pmatrix}s&0&-2b\\0&s&2a\\bs&-as&1-a^2-b^2\end{pmatrix} \tag{1.2.32.2}$$

Taking the common factor $s$ out of $C_1$ and $C_2$,

$$\left|\mathbf{M}\right|=s^2\begin{vmatrix}1&0&-2b\\0&1&2a\\b&-a&1-a^2-b^2\end{vmatrix} \tag{1.2.32.3}$$

The operation $R_3\to R_3-bR_1+aR_2$ is left multiplication by $\mathbf{E}_3=\begin{pmatrix}1&0&0\\0&1&0\\-b&a&1\end{pmatrix}$ with $\left|\mathbf{E}_3\right|=1$, and gives

$$\left|\mathbf{M}\right|=s^2\begin{vmatrix}1&0&-2b\\0&1&2a\\0&0&1+a^2+b^2\end{vmatrix}=s^2\cdot s=\left(1+a^2+b^2\right)^3 \tag{1.2.32.4}$$

which is the required result. The code verifies the steps symbolically.

For the alternate question, a point $(x,y)$ lies on the line through $A$ and $B$ iff, from (1.1.63.1),

$$\begin{vmatrix}x&y&1\\1&3&1\\0&0&1\end{vmatrix}=3x-y=0\implies y=3x \tag{1.2.32.5}$$

From (1.1.63.1), the area of $\triangle ABD$ is

$$\frac{1}{2}\left\|\begin{vmatrix}1&3&1\\0&0&1\\k&0&1\end{vmatrix}\right\|=\frac{1}{2}\left|3k\right|=3\implies k=\pm2 \tag{1.2.32.6}$$

codes/q32.py

1.2.33. Find the intervals in which the function $f$ defined as $f\left(x\right)=\sin x+\cos x$, $0\le x\le 2\pi$ is strictly increasing or decreasing.

**OR**

Prove that the radius of the right circular cylinder of greatest curved surface area which can be inscribed in a given cone is half that of the cone. (CBSE 65-C-3,

Class 12)

**Solution:** We solve the alternate question. Let the cone have radius $R$ and height $H$, and let the inscribed cylinder have radius $r$ and height $h$. By similar triangles,

$$\frac{h}{H} = \frac{R-r}{R} \implies h = H\left(1 - \frac{r}{R}\right) \tag{1.2.33.1}$$

The curved surface area of the cylinder is

$$S(r) = 2\pi rh = 2\pi H\left(r - \frac{r^2}{R}\right), \quad 0 < r < R \tag{1.2.33.2}$$

From (1.1.64.1), differentiating,

$$\frac{dS}{dr} = 2\pi H\left(1 - \frac{2r}{R}\right) = 0 \implies r = \frac{R}{2} \tag{1.2.33.3}$$

$$\frac{d^2S}{dr^2} = -\frac{4\pi H}{R} < 0 \tag{1.2.33.4}$$

so $S$ is maximum at $r = \frac{R}{2}$, with $S_{\max} = \frac{\pi RH}{2}$. Hence the radius of the cylinder of greatest curved surface area is half the radius of the cone.

The C code scans $r$ over $(0, R)$ and also solves $\frac{dS}{dr} = 0$ by bisection for several cones; in every case $\frac{r}{R} = 0.5$ and $S\left(\frac{R}{2}\right) = \frac{\pi RH}{2}$.

codes/q33.c

1.2.34. Evaluate $\int_1^3 \left(x^2 + 1 + e^x\right) dx$ as the limit of sums. (CBSE 65-C-3, Class 12)

**Solution:** Here $a = 1$, $b = 3$ and $f(x) = x^2 + 1 + e^x$. From (1.1.65.1), with $h = \frac{2}{n}$ so that $nh = 2$,

$$\int_1^3 f(x)\, dx = \lim_{h\to 0}\ h\sum_{i=1}^{n} f(1+ih) \tag{1.2.34.1}$$

$$f(1+ih) = (1+ih)^2 + 1 + e\cdot e^{ih} = 2 + 2ih + i^2h^2 + e\cdot e^{ih} \tag{1.2.34.2}$$

Using the sums in (1.1.65.1) and the geometric series,

$$h\sum_{i=1}^{n} f(1+ih) = 2nh + h^2 n(n+1) + \frac{h^3 n(n+1)(2n+1)}{6} + e\cdot\frac{h\,e^h\left(e^{nh}-1\right)}{e^h - 1} \tag{1.2.34.3}$$

Since $nh = 2$, we have $h^2 n(n+1) = 2(2+h)$ and $h^3 n(n+1)(2n+1) = 2(2+h)(4+h)$. As $h \to 0$, $\frac{h}{e^h - 1} \to 1$ and $e^h \to 1$, so

$$\int_1^3 \left(x^2 + 1 + e^x\right) dx = 4 + 4 + \frac{8}{3} + e\left(e^2 - 1\right) = \frac{32}{3} + e^3 - e \tag{1.2.34.4}$$

The code adds up the sums for increasing $n$ and compares with this value.

codes/q34.py

1.2.35. If the area between the curves $x = y^2$ and $x = 4$ is divided into two equal parts by the line $x = a$, then find the value of $a$ using integration.

**OR**

Find $\int \frac{x}{(x-1)^2(x+2)}\,dx$. (CBSE 65-C-3, Class 12)

**Solution:** We solve the alternate question. From (1.1.66.1),

$$\frac{x}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2} \tag{1.2.35.1}$$

Multiplying out,

$$x = (A+C)\,x^2 + (A+B-2C)\,x + (-2A+2B+C) \tag{1.2.35.2}$$

Comparing the coefficients of $x^2$, $x$ and 1,

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & -2 \\ -2 & 2 & 1 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \implies \begin{pmatrix} A \\ B \\ C \end{pmatrix} = \begin{pmatrix} \frac{2}{9} \\ \frac{1}{3} \\ -\frac{2}{9} \end{pmatrix} \tag{1.2.35.3}$$

Hence

$$\int \frac{x\,dx}{(x-1)^2(x+2)} = \frac{2}{9}\int \frac{dx}{x-1} + \frac{1}{3}\int \frac{dx}{(x-1)^2} - \frac{2}{9}\int \frac{dx}{x+2} \tag{1.2.35.4}$$

$$= \frac{2}{9}\ln\left|\frac{x-1}{x+2}\right| - \frac{1}{3(x-1)} + c \tag{1.2.35.5}$$

The code finds the partial fractions and checks that the derivative of the answer gives back the integrand.

codes/q35.py

1.2.36. If $y = x^{\sin x} + \sin^{-1}\sqrt{x}$, then find $\frac{dy}{dx}$. (CBSE 65-C-3, Class 12)

**Solution:** Let $y = u + v$ with $u = x^{\sin x}$ and $v = \sin^{-1}\sqrt{x}$. From (1.1.67.1),

$$\ln u = \sin x \ln x \implies \frac{1}{u}\frac{du}{dx} = \cos x \ln x + \frac{\sin x}{x} \tag{1.2.36.1}$$

$$\implies \frac{du}{dx} = x^{\sin x}\left(\cos x \ln x + \frac{\sin x}{x}\right) \tag{1.2.36.2}$$

From (1.1.22.1) and (1.1.68.1),

$$\frac{dv}{dx} = \frac{1}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x-x^2}} \tag{1.2.36.3}$$

Adding,

$$\frac{dy}{dx} = x^{\sin x}\left(\cos x \ln x + \frac{\sin x}{x}\right) + \frac{1}{2\sqrt{x-x^2}} \tag{1.2.36.4}$$

The code compares this with the derivative found by sympy at $x = 0.5$.

codes/q36.py